



Customer Shopping Behavior — Executive Overview

Analysis of 3,900 transactions to surface spending patterns, top products, subscription dynamics, and actionable recommendations for retail and e-commerce teams.

Dataset Snapshot

Scope

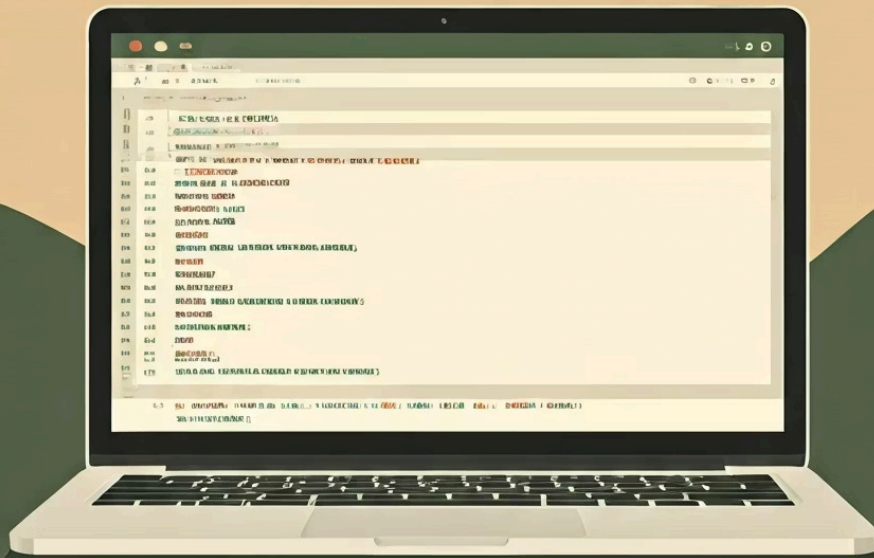
3,900 rows × 18 features. Key fields: demographics, item, category, price, season, discounts, frequency, rating, shipping, subscription.

Missing Data

37 missing values in Review Rating — imputed using category median to preserve product-level signal.

Data Quality

Columns standardized to snake_case. Dropped redundant promo_code_used after consistency checks with discount_applied.





Key Preparation & Feature Engineering

Performed null-handling, categorical cleaning, and created derived features: `age_group` (binned), `purchase_frequency_days`, and normalized `purchase_amount`. Result: consistent dataset ready for SQL and BI workflows.

- ❏ Imputation strategy: median rating per product category; preserves relative product differences without skewing means.

Revenue & Gender — SQL Findings



Finding: Male customers generated higher total revenue (\$157,890) vs Female (\$75,191).

Implication: Consider targeted merchandising and messaging for high-value male segments while investigating conversion drivers for female shoppers.

High-Spending Discount Users

839 transactions where discounts were applied but purchase amounts remained above average. These customers respond to promotions yet maintain higher basket values — a strategic audience for margin-aware offers and upsell bundles.

Action

Design tiered discounts tied to minimum spend to protect margins.

Action

Introduce personalized cross-sell suggestions during checkout.



Top-Rated & Discount-Dependent Products



Gloves — Top Rating

Average rating 3.86.

Opportunity: spotlight in quality-focused campaigns.



Sandals

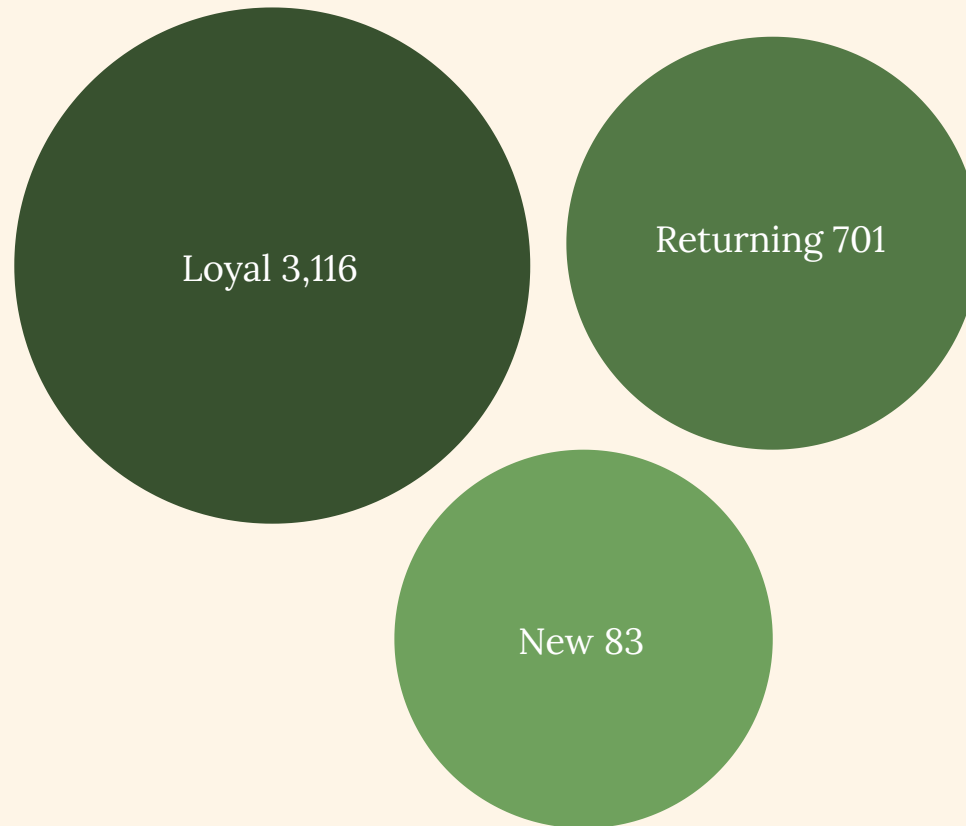
High rating and steady sales
— candidate for seasonal promotions.



Hat — Discount-Dependent

50% of purchases used discounts. Test full-price positioning and value bundles.

Customer Segments & Loyalty

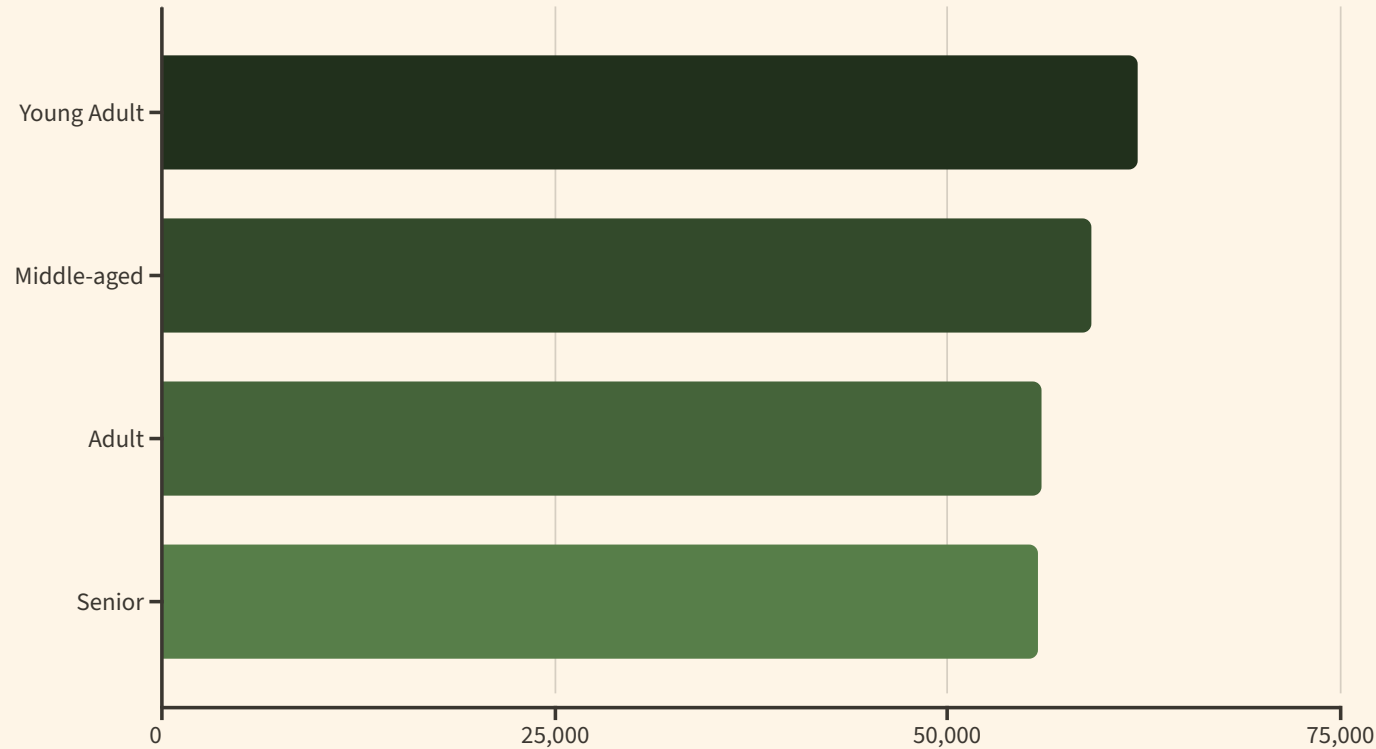


Segmentation: Loyal (3,116), Returning (701), New (83).

Insight: Majority are loyal repeat buyers — retention programs will have outsized ROI. New customer pipeline is small; focus acquisition efficiently.

- ❏ Consider cohort analysis to track retention by acquisition channel and onboarding flow.

Age Groups & Revenue — Priorities



Young adults contribute the largest revenue slice (~62K). Tailor product assortments and marketing to high-revenue groups while testing premium offers for middle-aged segments.

- Targeted promotions for Young Adults (mobile-first experiences)
- Premium bundles for Middle-aged customers



Dashboard Highlights (Power BI)

Interactive dashboard surfaced: subscription split (Yes 27% / No 73%), revenue & sales by category (Clothing leads), and age-group performance. KPIs: 3.9K customers, \$59.76 avg purchase, 3.75 avg rating.

- ❏ Use dashboard filters to isolate high-value cohorts (discount users, express-shipping) for campaign experiments.

Recommendations & Next Steps

Grow Subscriptions

Promote exclusive benefits (priority shipping, member-only bundles) to increase lifetime value of the 27% subscriber base.

Optimize Discounts

Introduce threshold-based discounts and margin-aware bundling for discount-prone products (hats, sneakers, coats).

Retention & Loyalty

Reward repeat buyers with tiered loyalty incentives; focus on converting Returning → Loyal segments.

Targeted Marketing

Allocate acquisition budget toward high-revenue age groups and test express-shipping messaging for conversion lift.

Next steps: run A/B tests for subscription offers, implement cohort tracking in BI, and model margin impact of revised discounting.